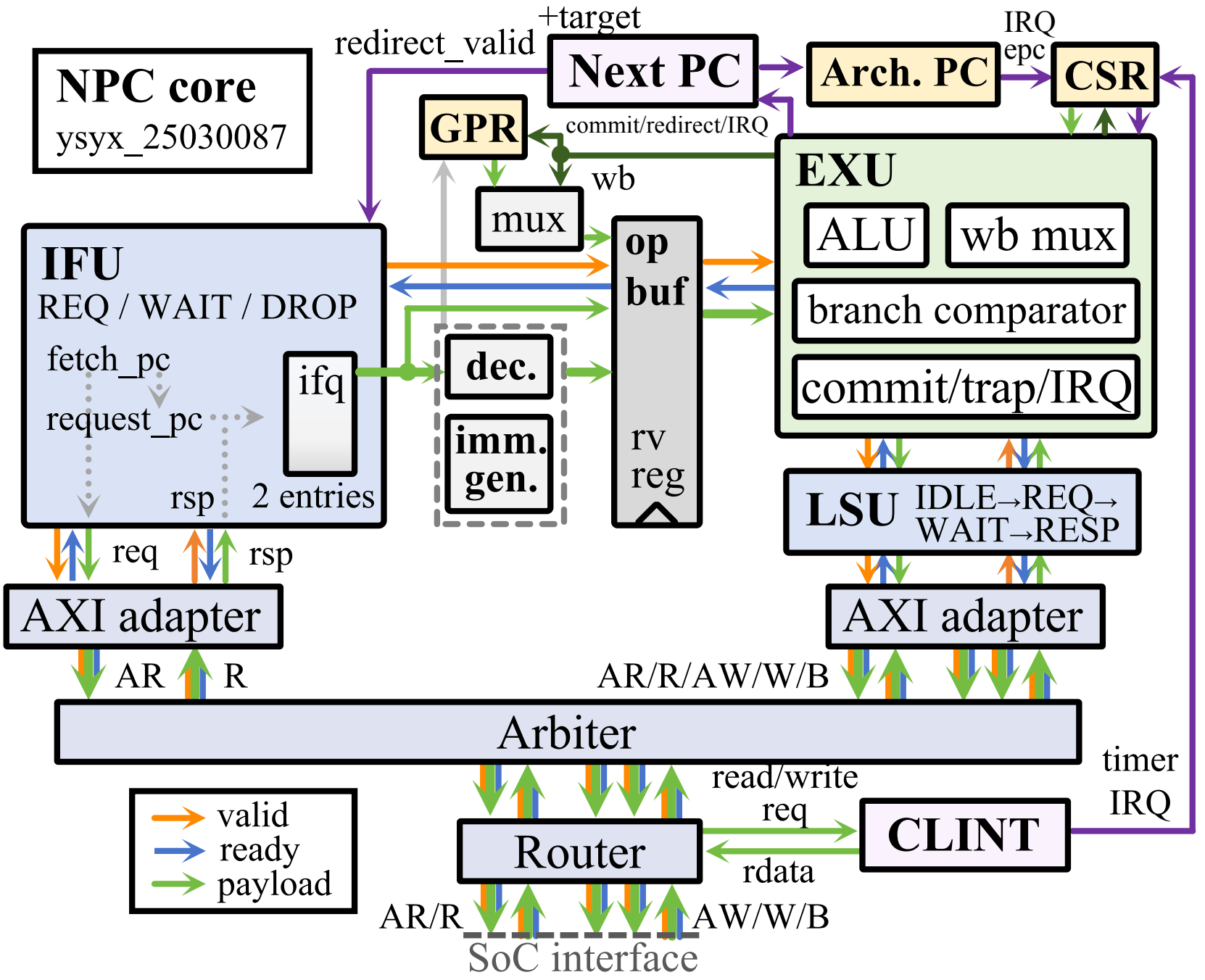

NPC core
ysyx_25030087
redirect_valid
+target
Next PC
Arch. PC
IRQ
epc
CSR
GPR
commit/redirect/IRQ
wb
mux
op
buf
EXU
ALU
wb mux
branch comparator
commit/trap/IRQ
IFU
REQ / WAIT / DROP
fetch_pc
request_pc
ifq
rsp
2 entries
dec.
imm.
gen.
rv
reg
LSU
IDLE→REQ→
WAIT→RESP
req
rsp
AXI adapter
AXI adapter
AR
R
AR/R/AW/W/B
Arbiter
timer
IRQ
read/write
req
rdata
CLINT
Router
AR/R
AW/W/B
SoC interface
valid
ready
payload